Department of Mechanical Engineering
Chittagong University of Engineering and Technology (C U E T)
Level 1, Term II, Examination 2023

Course No: Chem-133	Full Marks: 210	Time: 03 hours
Course Title: Chemistry of corrosion, environment and materials	Date: 02/ 12 / 2024	

The figures in the right margin indicate full marks. The questions are of equal value. There are 04 questions in each section. Answer any 03 questions from each section. Use separate script for each section.

Marks

Section: A

1. a) Draw a diagram of the atmospheric layers and label each layer's key features. What role does each layer play in protecting the earth? 15
   b) Why CO_2 and CH_4 are considered greenhouse gases that contribute to warming the planet, while O_2 and N_2 don't have the effect? 12
   c) What role does the aeration tank play in the process of treating wastewater? 8

2. a) Show the different steps for the disposal of hazardous waste. 15
   b) Classify solid wastes and discuss the characteristic of each category. 12
   c) What are the typical sources of wastewater from industries? 8

3. a) Explain the theory of electrochemical corrosion. How does the electrochemical series contribute to our understanding of electrochemical corrosion and the selection of materials in preventing galvanic corrosion? 15
   b) Explain the passivity of metal and stress corrosion. 12
   c) Discuss the various types of protective coatings used to prevent corrosion. 8

4. a) What is rusting and what chemical processes lead to the formation of rust on iron? 15
   b) How does the paints and varnishes protect the corrosion? 10
   c) What is stray current corrosion? How does it occur in metal structures? 10

Section: B

5. a) What are the main views of cement? Classify the Portland cement. 10
   b) Which criteria you have to keep in mind during manufacturing the cement with proper quality. 10
   c) What do you understand by setting and hardening of cement? Discuss the theories of setting and hardening with suitable chemical reactions? 10
   d) Make a comparative study between dry process and wet process. 5

6. a) What is the role of "plasticizer" in the modification of the polymeric and plastic compounds? 10
   b) Mention the reaction to produce polyvinyl Chloride (PVC). Write down essential constituents of PVC with their roles. 10
   c) Describe the structural differences relating to thermo-setting and thermo-plastic properties of polymer. 10
   d) What is vulcanization? Why is vulcanization so important to modify the rubber? 5

7. a) Define latex. What is the ratio of water and rubber hydrocarbon on it? What are the properties of rubber? 10
   b) Write down the ways by which you can vulcanize the rubber. How do you can indicate a vulcanized rubber? 15
   c) What are the properties of Neoprene rubber? Define Reclaimed rubber with example and merits. 10

8. a) "Ceramics are more stable than Glass"- Justify the statement. 10
   b) Design a process to produce 1250 tones molten glass for a glass industry. 10
   c) What are the effect of water pollution in the different parts of the environment? 10
   d) How does "Annealing Temperature" effect the quality of glass-Explain. 5

**********THE END**********